

**Topic 6**

Mensuration

**Concept guide · Worked examples · Practice · Investigation and modelling
· Solutions**

Learning focus

Content explanations, original Cambridge-style practice and an investigation task tailored to International Mathematics 0607.

Topic map

6.1	Units and compound measures
6.2	Perimeter and area
6.3	Circles, sectors and arcs
6.4	Volumes and surface areas
6.5	Similar solids and practical problems
6.6	Models with changing dimensions

Core foundations and Extended applications are integrated. Practise exact methods, calculator work and the written reasoning required in investigations.

1. Units and area

Convert lengths before calculating. Area is measured in square units: $1 \text{ m}^2 = 10000 \text{ cm}^2$; volume is in cubic units: $1 \text{ m}^3 = 1000000 \text{ cm}^3$. A triangle has area $\frac{1}{2} \times \text{base} \times \text{perpendicular height}$, and a trapezium has area $\frac{1}{2} \times \text{sum of parallel sides} \times \text{height}$.

Worked example

For a trapezium with parallel sides 7 cm and 11 cm and height 4 cm, area = $(7 + 11) \times \frac{4}{2} = 36 \text{ cm}^2$.

Exam check: The sloping edge of a triangle is not its perpendicular height unless it forms a right angle to the base.

2. Circles, arcs and sectors

The circle circumference is $2\pi r$ and area is πr^2 . A sector with central angle θ takes fraction $\frac{\theta}{360}$ of a full circle, so its arc length is $(\frac{\theta}{360})2\pi r$ and area is $(\frac{\theta}{360})\pi r^2$. Include straight radii when asked for sector perimeter.

Worked example

Radius 6 cm, angle 60° : arc = $(\frac{60}{360})2\pi(6) = 2\pi \text{ cm}$; sector area = $(\frac{60}{360})\pi(6^2) = 6\pi \text{ cm}^2$.

Exam check: Distinguish the curved arc from the full perimeter of a sector.

3. Prisms and cylinders

For a prism, volume = cross-sectional area $\times$ length. For a cylinder, volume = $\pi r^2 h$ and curved surface area = $2\pi r h$. Total closed surface area includes both circular ends; distinguish an open container from a closed one.

Worked example

A cylinder with $r = 3 \text{ cm}$ and $h = 8 \text{ cm}$ has volume $\pi(3^2)(8) = 72\pi \text{ cm}^3$.

Exam check: A diameter of 6 cm means $r = 3 \text{ cm}$; do not use 6 in πr^2 .

4. Cones, spheres and pyramids

A cone has volume $(\frac{1}{3})\pi r^2 h$ and curved surface area $\pi r l$, where l is slant height. A sphere has volume $(\frac{4}{3})\pi r^3$ and surface area $4\pi r^2$. A pyramid has volume $(\frac{1}{3}) \times \text{base area} \times \text{perpendicular height}$. Pythagoras often connects a cone height, radius and slant height.

Worked example

For a cone with $r = 3 \text{ cm}$ and $h = 4 \text{ cm}$, $l = 5 \text{ cm}$, so curved area = $15\pi \text{ cm}^2$ and volume = $12\pi \text{ cm}^3$.

Exam check: Use the perpendicular height for volume and slant height for curved surface area.

5. Composite and similar solids

Break a compound solid into standard parts. Add volumes of joined solids; subtract removed hollows. If lengths are multiplied by k , areas multiply by k^2 and volumes by k^3 . For material cost, use the actual exposed surface, including or excluding joined faces as appropriate.

Worked example

Two similar solids have length ratio 2:5. Their volume ratio is $2^3:5^3 = 8:125$.

Exam check: For a joined solid, an internal contact face is not part of the external surface.

6. Models with changing dimensions

When one dimension varies, write an expression before making a table. A fixed radius cylinder has volume linear in height, whereas a sphere has volume proportional to the cube of its radius. State which dimensions remain fixed.

Worked example

For $r=2$ cm, cylinder volume $V(h)=\pi(2^2)h=4\pi h$ cm³.

Exam check: A graph of volume against h is a line only if radius stays constant.

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Graphical calculator and Paper 6

Keep π exact until the end when appropriate. Use a table to explore how volume changes as one dimension varies.

State the rule or model, display enough working to reproduce your result, test it against the given values and discuss at least one limitation.



Practice paper

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Instructions

Original practice, informed by June 2026 components. Section A: short multiple choice for revision without a calculator. Section B: show full written working. Section C: investigation and modelling with reasons and checks.

Candidate name:

Class / date:

Total: 40 marks

Section A · Multiple choice

Choose one answer for each question.

1. 1 m^2 equals how many cm^2 ? [1]

- A 100
- B 1000
- C 10000
- D 1000000

2. Area of a triangle with base 8 cm and height 5 cm? [1]

- A 13
- B 20
- C 40
- D 80

3. Circumference of a circle with radius 4? [1]

- A 4π
- B 8π
- C 16π
- D 32π

4. Volume of a cylinder with radius 2 and height 5? [1]

- A 10π
- B 20π
- C 25π
- D 40π

5. Area of a 90° sector with radius 4? [1]

- A 2π
- B 4π
- C 8π
- D 16π

6. Volume of a cone with radius 3 and height 6? [1]

- A 6π
- B 18π
- C 27π
- D 54π

7. Surface area of a sphere with radius 2? [1]

- A 8π
- B 12π
- C 16π
- D 32π

8. Length scale factor 4 gives volume factor [1]

- A 4
- B 8
- C 12
- D 64

Section B · Structured questions

9. A sector has radius 8 cm and central angle 90° .

(a) Find its arc length in terms of π . [2]

(b) Find its area in terms of π . [2]

(c) Find its perimeter in terms of π . [2]

[Total: 6]

10. A closed cylinder has radius 3 cm and height 10 cm.

(a) Calculate its volume in terms of π . [2]

(b) Calculate its curved surface area in terms of π . [2]

(c) Calculate its total surface area in terms of π .

[2]

[Total: 6]

11. A cone has radius 5 cm and perpendicular height 12 cm.

(a) Find its slant height.

[2]

(b) Find its volume in terms of π .

[2]

(c) Find its curved surface area in terms of π .

[2]

[Total: 6]

12. Two similar cylinders have radii in the ratio 3:5. The smaller has volume 54 cm^3 and surface area 72 cm^2 .

(a) Find their volume ratio.

[2]

(b) Find the larger volume.

[2]

(c) Find the larger surface area.

[2]

[Total: 6]

Section C · Investigation and modelling

Show the rule, test it with values and explain the result in context.

13. A cylindrical tank has fixed radius 2 m and variable height h m.

(a) Write its volume V in terms of h . [2]

(b) Calculate V for $h=3$ m. [2]

(c) Find h when $V=40\pi$ m³. [2]

(d) State one practical constraint a tank designer would add to this model. [2]

[Total: 8]

Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Method, model, communication and verification matter. A numerical answer should follow from clear working.

Section A · Answers and reasons

1. C — 10000. $1\text{ m} = 100\text{ cm}$; square the factor to obtain 10000.
2. B — 20. Triangle area is $\frac{1}{2} \times \text{base} \times \text{perpendicular height}$.
3. B — 8π . Circumference $= 2\pi r = 8\pi$.
4. B — 20π . Volume $= \pi r^2 h = \pi(2^2)(5) = 20\pi$.
5. B — 4π . Quarter-circle area $= \pi(4^2)/4 = 4\pi$.
6. B — 18π . Cone volume $= \pi r^2 h/3 = \pi(9)(6)/3$.
7. C — 16π . Sphere area $= 4\pi r^2 = 4\pi(4)$.
8. D — 64. Volume factor is the cube of the length factor: 4^3 .

Section B • Worked solutions

9. a) $(90/360) \times 2\pi(8) = 4\pi$ cm [2]. b) $(90/360) \times \pi(8^2) = 16\pi$ cm² [2]. c) $4\pi + 8 + 8 = 16 + 4\pi$ cm [2].

10. a) $\pi(3^2)(10) = 90\pi$ cm³ [2]. b) $2\pi(3)(10) = 60\pi$ cm² [2]. c) $60\pi + 2\pi(3^2) = 78\pi$ cm² [2].

11. a) $l = \sqrt{5^2 + 12^2} = 13$ cm [2]. b) $(1/3)\pi(5^2)(12) = 100\pi$ cm³ [2]. c) $\pi(5)(13) = 65\pi$ cm² [2].

12. a) $3^3:5^3 = 27:125$ [2]. b) $54 \times 125/27 = 250$ cm³ [2]. c) $72 \times 25/9 = 200$ cm² [2].

Section C • Investigation and modelling solutions

13(a). $V = \pi r^2 h = 4\pi h$ m³ [2].

13(b). 12π m³ [2].

13(c). $40\pi = 4\pi h$, so $h = 10$ m [2].

13(d). Maximum building height, material strength or capacity imposes a limit [2].

Self-check

Explain why your model fits the starting data, show an independent check and name a condition under which your prediction would be unreliable.

Reference: Cambridge IGCSE International Mathematics 0607 syllabus 2025-2027, <https://www.cambridgeinternational.org/Images/662472-2025-2027-syllabus.pdf>. Assessment conventions informed by the June 2026 papers and mark schemes supplied by the user. All practice questions are newly written.